

PCHM — Prediction Change Monitoring Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

PMRS — Prediction Monitoring & Revision Standard

Operational Layer: Prediction Monitoring & Revision

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Prediction Change Monitoring

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

PCHM governs the monitoring of changes capable of materially affecting an existing Governed Prediction after formation.

It establishes how relevant predictive conditions are observed, compared with their governed reference state, classified, and preserved so that potentially material change can be identified before the prediction becomes stale, misleading, or operationally disconnected from the conditions under which it was formed.

PCHM governs the transition:

Current Governed Prediction + Governed Reference Conditions + Observed Change Signals → Governed Prediction Change Monitoring State

It does not decide whether the prediction must be revised.



It establishes whether relevant change exists and whether that change requires further governance.

Canonical Definition

The Prediction Change Monitoring Module (PCHM) defines the operational governance requirements by which changes in the conditions surrounding a Governed Prediction are systematically observed, associated with the prediction, compared with the relevant governed reference state, classified for potential predictive relevance, and preserved for subsequent Review Trigger determination.

PCHM monitors changes affecting or potentially affecting:

Prediction Basis; Predictive Assumptions; probability-relevant conditions; uncertainty; confidence-relevant conditions; feasibility; collective predictive structure; Predictor–Reality Proximity; operational environment; external environment; prediction use; Prediction Horizon; prediction-state dependencies.

PCHM does not treat every observed change as material.

Its purpose is to make relevant change governable before decisions about review or revision are made.

A prediction cannot be responsibly maintained if the conditions capable of changing its meaning are not observable.

Operational Role

PCHM is the first operational module within PMRS.

Its focused operational question is:

What has changed since the prediction's current governed state was established, and could that change materially affect the prediction or the conditions under which it remains legitimate?

PCHM creates the monitored change state required by subsequent PMRS governance.

It does not:

recalculate the prediction; revise probability; determine that review is mandatory; revise the prediction; supersede the prediction; withdraw the prediction.

Its role is:

Observe → Associate → Compare → Classify → Preserve → Escalate Potentially Relevant Change

Internal Module Flow

Prediction Change Monitoring → Review Trigger → Prediction Revision → Prediction Supersession → Prediction Withdrawal

Current Position:

Prediction Change Monitoring

1. Prediction Monitoring Object

Every monitoring process must identify the prediction being monitored.

The Prediction Monitoring Object should remain associated with:

Prediction ID; current prediction version; Prediction Object; Predictive Purpose; Prediction Horizon; Predictive Boundary; current prediction state.

Monitoring Without Prediction Identity ≠ Governed Prediction Monitoring

2. Monitoring Reference State

Change can be determined only relative to a reference state.

PCHM therefore requires a governed Monitoring Reference State representing relevant conditions associated with the prediction when it was:

formed; last materially revised; last formally reviewed; otherwise established as current.

The reference state provides the baseline against which subsequent conditions are compared.

Observed State – Reference State → Potential Predictive Change

3. Monitoring Scope

Monitoring Scope defines which conditions are sufficiently relevant to the prediction to justify observation.

Monitoring Scope may include:

Prediction Basis; assumptions; source conditions; operational conditions; market conditions; infrastructure conditions; regulatory conditions; technological conditions; environmental conditions; collective prediction structure; prediction use; Predictor–Reality Proximity; relevant external events.

Monitoring Scope should remain proportionate to the prediction.

Everything That Changes ≠ Relevant Predictive Change

4. Monitoring Proportionality

Not every prediction requires identical monitoring intensity.

Monitoring should reflect:

Prediction Horizon; volatility; uncertainty; assumption sensitivity; consequence; reliance; use; operational criticality; environmental change rate.

A low-consequence long-term exploratory prediction may require periodic monitoring.

A high-consequence operational prediction may require continuous monitoring.

5. Monitoring Mode

PCHM supports multiple monitoring modes.

Continuous Monitoring

Relevant conditions are observed continuously or near continuously.

Periodic Monitoring

Conditions are reviewed according to a defined interval.

Event-Driven Monitoring

Specific events initiate monitoring activity.

Threshold-Driven Monitoring

Monitoring responds when defined values cross relevant thresholds.

Condition-Driven Monitoring

Monitoring activates when a defined predictive condition changes.

Manual Monitoring

Authorized human or organizational actors initiate monitoring.

Automated Monitoring

Systems detect and classify relevant change signals.

No single monitoring mode is universally required.

6. Monitoring Frequency

Monitoring Frequency should reflect the speed at which relevant predictive conditions can materially change.

Possible frequencies may include:

real-time; hourly; daily; weekly; monthly; milestone-based; event-based.

Higher Monitoring Frequency $\neq$ Better Governance by Default

Excessive monitoring can generate noise without improving predictive governance.

7. Change Signal

A Change Signal is information indicating that a condition relevant to the prediction may have changed.

Examples include:

new data; new observation; changed source;

assumption deterioration; policy announcement; operational disruption; market movement; infrastructure failure; new technological capability; collective participant change; new use pattern.

A signal does not establish materiality.

Change Signal $\neq$ Material Change

8. Signal Source

Every material Change Signal should remain attributable to an identifiable source where possible.

The source may be:

sensor; dataset; human report; institution; market; AI system; external database; operational system; public authority; governed evidence source.

PCHM does not establish evidence validity itself.

Where required, ADIT® may govern evidence.

9. Signal Relevance

PCHM determines whether the signal relates to a governed predictive condition.

A signal may be:

Directly Relevant

Potentially Relevant

Indirectly Relevant

Not Materially Relevant

Unable to Determine

Signal relevance is distinct from signal materiality.

10. Change Observation

A Change Observation records an observed difference between current and reference conditions.

The observation should distinguish:

What Was Previously Known

from:

What Is Currently Observed

This enables subsequent change assessment.

11. Change Association

Observed change must be associated with the predictive structure it may affect.

Possible associations include:

Prediction Basis; assumption; probability; uncertainty; confidence; feasibility; collective structure; Prediction Horizon; Predictive Boundary; prediction use; operational condition.

Observed Change Without Predictive Association $\neq$ Governed Change Determination

12. Prediction Basis Monitoring

PCHM monitors whether information forming the Prediction Basis remains materially comparable with the current environment.

Relevant changes may include:

source updates; source removal; contradictory information; new information; source deterioration; data revision; changing signal relevance.

PBAS continues to own Prediction Basis governance.

PCHM owns monitoring of post-formation changes affecting that basis.

13. Prediction Basis Decay Monitoring

Prediction Basis Decay may develop gradually.

PCHM should be capable of detecting:

data ageing; source staleness; declining contextual relevance; source degradation; growing contradiction; environment drift.

Prediction Basis Age $\neq$ Prediction Basis Decay by Default

Old information may remain relevant.

Recent information may already be obsolete.

14. Data Change

New data may:

reinforce the prediction; weaken the prediction; contradict the prediction; remain neutral; require further analysis.

PCHM records the change.

It does not independently determine the revised prediction.

15. Source Change

A source may materially change even where its name remains constant.

Examples:

methodology changes; ownership changes; measurement changes; collection procedures change; update frequency changes; source scope changes.

Same Source Name $\neq$ Same Source Condition

16. Source Loss

A prediction may depend upon a source that becomes unavailable.

Source Loss may affect:

monitoring capability; Prediction Basis continuity; assumption support; current predictive legitimacy.

Source Loss should remain visible even where no replacement source exists.

17. Source Replacement

Replacing a source does not automatically preserve predictive equivalence.

A replacement source may differ in:

methodology;

scope; reliability; frequency; population; geography; measurement logic.

Replacement Source $\neq$ Equivalent Source by Default

18. Assumption Monitoring

PCHM monitors conditions supporting material Predictive Assumptions.

A monitored assumption may remain:

Supported

Potentially Weakened

Materially Weakened

Potentially Failed

Unable to Determine

PAS governs the assumption itself.

PCHM governs monitoring of its continuing condition.

19. Assumption Dependency Monitoring

Some assumptions support multiple parts of the prediction.

A single assumption change may therefore propagate across:

probability; feasibility; Prediction Basis interpretation; Prediction Horizon; use.

PCHM should preserve such dependencies.

20. Assumption Sensitivity and Monitoring Priority

Assumptions with greater predictive sensitivity may justify greater monitoring priority.

High Assumption Sensitivity $\rightarrow$ Potentially Higher Monitoring Importance

This does not automatically determine monitoring frequency.

21. Probability-Relevant Change

PCHM may detect information capable of materially affecting predictive probability.

It does not calculate the new probability.

The proper transition is:

Probability-Relevant Change Detected $\rightarrow$ Further Governance Required

not:

Change Detected → Probability Automatically Changed

22. Uncertainty Change Monitoring

Uncertainty may change because:

information becomes incomplete; source disagreement increases; conditions become volatile; assumptions weaken; model applicability changes.

PCHM should identify material uncertainty-relevant changes even where the headline probability remains unchanged.

23. Confidence-Relevant Change

Confidence-relevant conditions may change independently from probability.

Examples:

weaker data quality; deteriorating methodology applicability; changed environment; increased source disagreement.

Stable Probability ≠ Stable Confidence

24. Range-Relevant Change

Where predictions include ranges, new conditions may affect:

range width; lower boundary; upper boundary; distribution shape; conditional ranges.

PCHM monitors relevant changes without independently reforming the range.

25. Conditionality Monitoring

Conditional predictions depend upon explicit conditions.

PCHM should monitor whether those conditions remain:

Satisfied

Partially Satisfied

At Risk

No Longer Satisfied

Unable to Determine

Conditional Prediction ≠ Permanently Valid Conditional Structure

26. Feasibility Monitoring

PCHM monitors conditions capable of changing Prediction Feasibility.

Examples:

resource availability; technical capability; infrastructure; regulatory permission; operational capacity; economic viability.

PFES determines feasibility.

PCHM identifies changes requiring potential reassessment.

27. Constraint Change

A new constraint may materially alter prediction feasibility or probability.

Constraints may be:

physical; technical; financial; regulatory; organizational; temporal; infrastructural.

Constraint changes should be associated with the relevant predictive condition.

28. Opportunity Change

Not all relevant changes are negative.

New capabilities may increase feasibility or probability.

Examples:

new technology; new funding; infrastructure completion; regulatory approval; additional capacity.

PCHM monitors positive and negative changes symmetrically.

29. Operational Condition Monitoring

Operational conditions may materially affect predictions.

Examples include:

production capacity; system availability; staffing; logistics; resource availability; infrastructure status.

ORA™ may establish relevant Operational Reality.

PCHM consumes such governed information where appropriate.

30. External Condition Monitoring

External predictive conditions may include:

economic environment; regulation; geopolitics; climate; technology; market structure; supply environment; public behavior.

PCHM monitors only conditions relevant to the governed prediction.

31. Intervening Event Detection

An Intervening Event occurs after prediction formation and may alter the predicted environment.

Examples:

regulatory decision; acquisition; conflict; natural disaster; system failure; technological breakthrough; major policy change.

Intervening Event $\neq$ Material Predictive Change by Default

Materiality belongs to subsequent governance.

32. Event Timing

The timing of an event relative to:

prediction formation; last revision;

Prediction Horizon; decision use;

may affect its predictive relevance.

PCHM therefore preserves event timing.

33. Event Scope

An event may affect:

entire Prediction Object; one condition; one geography; one population; one time period; one assumption.

PCHM should avoid treating local change as universal change without support.

34. Prediction Horizon Monitoring

PCHM monitors the relationship between current time and Prediction Horizon.

Monitoring may intensify as the horizon approaches where relevant.

However:

Approaching Prediction Horizon $\neq$ Automatic Revision Requirement

35. Temporal Drift

Temporal Drift occurs where the environment gradually diverges from the temporal conditions represented in the prediction.

This may occur without a single identifiable event.

PCHM should support detection of gradual as well as abrupt change.

36. Environmental Drift

Environmental Drift occurs where relevant predictive conditions progressively move outside the environment represented by the current prediction.

Examples include:

market regime change; technology adoption; climate shift; organizational transformation; infrastructure change.

Gradual Change $\neq$ Non-Material Change

37. Structural Change

Structural Change alters the underlying system rather than merely changing a temporary value.

Examples:

new regulatory regime; market redesign; organizational restructuring; infrastructure replacement; technological transition.

Structural change may justify elevated monitoring significance.

38. Collective Prediction Monitoring

Where the prediction is collective, PCHM may monitor:

participant composition; source diversity; dependence; aggregation conditions; influence concentration; participation concentration; collective representation state.

CPGS retains ownership of Collective Prediction governance.

39. Collective Composition Change

The same collective prediction value may arise from a materially different participant structure.

Example:

72% with 10,000 diverse participants

versus:

72% with 80% of effective influence concentrated in three actors

Therefore:

Same Collective Number $\neq$ Same Collective Structure

40. Predictive Dependence Change

Predictors previously considered independent may begin using:

common data; common models; common research; common upstream sources.

This may materially reduce effective predictive diversity.

41. Influence Concentration Change

A market or collective may become increasingly dominated by a small number of participants.

PCHM should preserve changes in influence concentration where relevant.

Stable Aggregate $\neq$ Stable Influence Structure

42. Predictor–Reality Proximity Monitoring

Predictor–Reality Proximity may change after prediction formation.

A predictor may become:

more information-advantaged; decision-proximate; outcome-informed; outcome-influencing; outcome-controlling.

PCHM records material transitions.

It does not infer misconduct.

43. Predictor–Reality Proximity Boundary

High Predictor–Reality Proximity $\neq$ Misconduct

Decision Proximity $\neq$ Decision Authority

Outcome Knowledge $\neq$ Predictive Capability

PCHM monitors the relationship.

It does not assign guilt, intent, or legal responsibility.

44. Prediction Use Monitoring

PCRS governs Prediction Use Boundary.

PCHM may monitor whether actual use changes conditions relevant to the prediction.

This is particularly important where prediction use can alter Operational Reality.

45. Prediction Feedback Signal

A Prediction Feedback Signal indicates that communication, reliance, or use of the prediction may have influenced the environment.

Example:

Demand Prediction $\rightarrow$ Inventory Expansion $\rightarrow$ Market Supply Change

Such feedback may become relevant to continuing predictive legitimacy.

46. Reflexive Change Monitoring

Reflexive predictions require monitoring of whether prediction dissemination or use affects the predicted system.

Canonical relationship:

Prediction → Communication → Reliance → Action → Changed Operational Reality

PCHM monitors the change.

It does not determine final causal attribution.

47. Self-Fulfilling Dynamics

Prediction use may increase the probability of the predicted outcome.

Example:

Shortage Prediction → Increased Purchasing → Shortage

PCHM should preserve relevant feedback conditions.

48. Self-Negating Dynamics

Prediction use may reduce the probability of the predicted outcome.

Example:

Failure Prediction → Preventive Maintenance → Failure Avoided

The changed condition should remain identifiable.

Prevented Outcome ≠ Automatically Failed Prediction

49. Prediction Communication Change

A prediction may be communicated differently without the underlying prediction changing.

PCHM should distinguish:

Prediction Change

from:

Representation Change

PRSM governs representation.

Communication Change ≠ Prediction Change

50. Reliance Change

A prediction may become more or less relied upon while its predictive content remains unchanged.

RLCM governs reliance.

PCHM may monitor reliance changes where they affect the predictive environment.

Reliance Change ≠ Prediction Change

51. Use Boundary Change

PUBM may alter permitted use without altering the prediction itself.

PCHM should preserve this distinction.

Use Boundary Change ≠ Prediction Revision

52. Monitoring Noise

Monitoring systems may produce large volumes of irrelevant variation.

PCHM should prevent:

More Signals

from automatically becoming:

More Governance Events

Noise should remain distinguishable from potentially meaningful change.

53. Duplicate Signals

The same underlying event may appear through many sources.

PCHM should avoid treating duplicate reporting as multiple independent changes.

Multiple Reports $\neq$ Multiple Changes

54. Correlated Signals

Different signals may share the same origin.

Where relevant, their dependence should remain identifiable.

Multiple Signals $\neq$ Independent Evidence of Change

55. Conflicting Signals

Monitoring may detect contradictory information.

Conflicting signals should remain visible rather than being silently averaged or suppressed.

Possible state:

Change Condition Uncertain

This may itself justify further assessment.

56. Missing Signal

Absence of expected monitoring information may itself be relevant.

Examples:

sensor stops reporting; dataset update fails; source disappears; monitoring system becomes unavailable.

No New Signal $\neq$ No Change by Default

57. Monitoring Blind Spot

A Monitoring Blind Spot exists where a condition capable of materially affecting the prediction is not sufficiently observable.

Blind spots should be identifiable.

Unobserved Condition $\neq$ Stable Condition

58. Monitoring Coverage

Monitoring Coverage represents the extent to which relevant predictive conditions are actually observable.

Coverage may be:

Sufficient

Partially Sufficient

Material Gap

Insufficient

Unable to Determine

59. Change Persistence

A change may be:

Transient

Recurring

Persistent

Structural

Persistence may affect its potential predictive relevance.

PCHM records persistence without making the final Review Trigger determination.

60. Change Direction

Where meaningful, PCHM may identify whether a change:

Supports Existing Prediction

Weakens Existing Prediction

Contradicts Existing Prediction

Changes Conditions Without Clear Direction

Unable to Determine

This is not a revised probability.

61. Change Magnitude

Change Magnitude may be classified where appropriate.

Possible states:

Minor

Moderate

Substantial

Major

Unable to Determine

Magnitude alone does not establish predictive materiality.

62. Change Velocity

The speed of change may matter.

A slowly changing condition and a rapidly deteriorating condition may require different monitoring responses.

Possible states:

Stable

Slow

Accelerating

Rapid

Abrupt

63. Change Scope

PCHM may identify whether a change affects:

one prediction component; multiple components;
entire predictive environment.

Scope helps downstream Review Trigger governance determine significance.

64. Change Reversibility

Some changes may quickly reverse.

Others may be effectively permanent.

Change reversibility may therefore be relevant to monitoring classification.

Temporary Change $\neq$ Irrelevant Change

65. Change Interaction

Several individually limited changes may combine into a material predictive condition.

Example:

slightly weaker source + slightly weaker assumption + slightly reduced feasibility
may collectively become significant.

PCHM should preserve interaction where identifiable.

Individually Minor Changes $\neq$ Collectively Minor Change

66. Cumulative Change

Gradual accumulation may materially alter the predictive environment without any single dramatic event.

PCHM should support cumulative monitoring.

This protects against slow prediction staleness.

67. Monitoring State

PCHM may classify the current monitoring state as:

Stable

Change Observed

Potentially Material Change

Multiple Relevant Changes

Monitoring Intensification Required

Monitoring Coverage Insufficient

Change Condition Uncertain

Unable to Determine

These are monitoring states.

They are not Prediction Review or Revision states.

68. Change Escalation Candidate

Where observed change may be material, PCHM produces a:

Change Escalation Candidate

This becomes input to:

RTM — Review Trigger Module

PCHM does not determine the final trigger.

69. Monitoring Intensification

Monitoring may be intensified where:

signals conflict; change accelerates; assumption stability deteriorates; major event occurs; Prediction Horizon approaches; use becomes more consequential; monitoring coverage becomes uncertain.

Monitoring Intensification $\neq$ Prediction Revision

70. Monitoring De-Intensification

Monitoring may also be reduced where conditions become sufficiently stable and consequence remains appropriate.

Monitoring governance should remain proportional rather than permanently escalating.

71. Monitoring Failure

Monitoring Failure may occur where:

required conditions were not observed; monitoring stopped; signal ingestion failed; relevant change was missed; stale sources continued without detection.

Monitoring Failure $\neq$ Prediction Error

72. Monitoring Integrity

Monitoring should preserve the difference between:

observed fact; inferred change; potential predictive relevance; materiality determination.

These should not be collapsed into one state.

73. Monitoring Traceability

Material monitoring records should permit reconstruction of:

Prediction State $\rightarrow$ Reference Condition $\rightarrow$ Observed Signal $\rightarrow$ Change Observation $\rightarrow$ Predictive Association $\rightarrow$ Monitoring State

This creates traceability without prematurely rewriting the prediction.

74. Monitoring Timestamp

Monitoring records should preserve when:

signal occurred; signal was observed; change was recorded; escalation occurred.

These times may differ.

Event Time $\neq$ Detection Time

75. Historical Monitoring Preservation

Later information must not rewrite what the monitoring system actually knew at an earlier time.

Later Knowledge $\neq$ Earlier Monitoring State

This is necessary for later lifecycle reconstruction.

76. Machine-Readable Monitoring

PCHM should support machine-readable monitoring structures where appropriate.

A monitoring record may include:

Prediction ID; prediction version; monitored condition; reference state; current observation; signal source; event time; detection time; relevance state; magnitude; direction; persistence; monitoring state; escalation candidate.

Machine-Readable Monitoring $\neq$ Autonomous Revision Authority

77. Human Monitoring

Human experts may identify relevant change that automated systems fail to detect.

Human observations should remain distinguishable from validated predictive changes.

Human Observation $\neq$ Automatic Materiality Determination

78. AI-Assisted Monitoring

AI systems may detect:

anomalies; source changes; assumption deterioration; conflicting information; environmental drift; emerging events.

AI detection may support PCHM.

But:

AI-Detected Change $\neq$ Governed Revision

79. Autonomous Monitoring

Autonomous systems may continuously monitor predictive conditions.

Their monitoring authority does not automatically include authority to revise the prediction.

Monitoring Authority $\neq$ Revision Authority

80. Cross-System Monitoring

A prediction may depend upon conditions observed across multiple systems.

PCHM should support association of distributed Change Signals with one governed prediction.

Technical distribution must not break predictive traceability.

81. Governance Requirements

A governed PCHM implementation should be able to:

1. identify the Governed Prediction being monitored;
2. identify the current prediction version;
3. establish a Monitoring Reference State;
4. define Monitoring Scope;
5. establish proportionate monitoring intensity;
6. define applicable Monitoring Mode;
7. define Monitoring Frequency where relevant;
8. identify Change Signals;
9. identify signal sources;
10. assess signal relevance;
11. record Change Observations;
12. associate change with predictive structure;
13. monitor Prediction Basis change;
14. monitor Prediction Basis Decay;
15. monitor data change;
16. monitor source change;
17. identify Source Loss;
18. identify Source Replacement;
19. preserve source non-equivalence where relevant;
20. monitor Predictive Assumptions;
21. monitor assumption dependencies;
22. consider assumption sensitivity in monitoring priority;
23. identify probability-relevant change;
24. monitor uncertainty-relevant change;
25. monitor confidence-relevant

change; 26. monitor range-relevant change; 27. monitor predictive conditionality; 28. monitor feasibility-relevant conditions; 29. identify Constraint Change; 30. identify Opportunity Change; 31. monitor relevant Operational Conditions; 32. monitor relevant External Conditions; 33. detect Intervening Events; 34. preserve Event Timing; 35. preserve Event Scope; 36. monitor Prediction Horizon proximity; 37. detect Temporal Drift; 38. detect Environmental Drift; 39. identify Structural Change; 40. monitor Collective Prediction conditions where applicable; 41. identify Collective Composition Change; 42. identify Predictive Dependence Change; 43. identify Influence Concentration Change; 44. monitor Predictor–Reality Proximity where relevant; 45. preserve Predictor–Reality Proximity boundaries; 46. monitor Prediction Use feedback where relevant; 47. identify Prediction Feedback Signals; 48. monitor Reflexive Change; 49. preserve Self-Fulfilling Dynamics where relevant; 50. preserve Self-Negating Dynamics where relevant; 51. distinguish Prediction Change from Communication Change; 52. distinguish Prediction Change from Reliance Change; 53. distinguish Prediction Change from Use Boundary Change; 54. distinguish monitoring noise from relevant change; 55. identify duplicate signals; 56. identify correlated signals; 57. preserve conflicting signals; 58. identify Missing Signals; 59. identify Monitoring Blind Spots; 60. assess Monitoring Coverage; 61. identify Change Persistence; 62. identify Change Direction where supportable; 63. identify Change Magnitude where supportable; 64. identify Change Velocity; 65. identify Change Scope; 66. identify Change Reversibility where relevant; 67. identify Change Interaction; 68. identify Cumulative Change; 69. determine Monitoring State; 70. create Change Escalation Candidate where appropriate; 71. support Monitoring Intensification; 72. support Monitoring De-Intensification; 73. identify Monitoring Failure; 74. preserve Monitoring Integrity; 75. preserve Monitoring Traceability; 76. preserve monitoring timestamps; 77. preserve historical monitoring states; 78. support machine-readable monitoring; 79. support human monitoring; 80. support AI-assisted monitoring; 81. support autonomous monitoring without granting Revision Authority; 82. support cross-system monitoring; 83. prevent Change Detection from becoming automatic Revision Decision; 84. preserve potentially relevant change for RTM; 85. produce a Governed Prediction Change Monitoring State.

82. Governance Logic

PCHM applies the following operational logic:

What prediction and prediction version are being monitored?

→ establish the Prediction Monitoring Object.

Against what governed conditions should change be measured?

→ establish the Monitoring Reference State.

Which conditions are relevant enough to monitor?

→ establish Monitoring Scope.

Has a Change Signal been observed?

If no:

→ preserve current monitoring state and continue applicable monitoring.

If yes:

→ identify the signal source and observed change.

Can the change be associated with a governed predictive condition?

If no:

→ classify as not materially relevant or Unable to Determine.

If yes:

→ preserve the predictive association.

Does the change potentially affect Prediction Basis, assumptions, probability, uncertainty, confidence, feasibility, collective structure, Predictor–Reality Proximity, Prediction Horizon, Operational Conditions, or prediction-use feedback?

If yes:

→ classify its direction, magnitude, persistence, scope, and other relevant characteristics where supportable.

Could the observed change materially affect continuing predictive legitimacy?

If potentially yes:

→ create a Change Escalation Candidate .

Is Monitoring Coverage insufficient to determine the condition?

If yes:

→ classify Monitoring Coverage Insufficient or Unable to Determine and escalate where appropriate.

At no point does PCHM independently revise the prediction.

Change Detection → Potential Review Escalation

not:

Change Detection → Automatic Revision

83. Minimum Implementation Framework

A. Prediction Reference

Identifier of the Governed Prediction being monitored.

B. Prediction Version

Current prediction version associated with monitoring.

C. Monitoring Reference State

Governed baseline against which current conditions are compared.

D. Monitored Condition

Prediction Basis, assumption, feasibility condition, external condition, collective condition, or other relevant predictive dependency.

E. Change Signal

Observed information indicating possible change.

F. Signal Source

Origin of the Change Signal.

G. Event & Detection Time

When the relevant event occurred and when it was detected.

H. Predictive Association

Predictive component potentially affected by the change.

I. Change Direction

Supporting, weakening, contradicting, neutral/unclear, or Unable to Determine.

J. Change Magnitude

Minor, moderate, substantial, major, or Unable to Determine.

K. Change Persistence & Velocity

Transient, recurring, persistent, structural and applicable rate of change.

L. Monitoring Coverage

Sufficient, partially sufficient, material gap, insufficient, or Unable to Determine.

M. Monitoring State

Current PCHM monitoring classification.

N. Change Escalation Candidate

Whether the observed condition should proceed to RTM for Review Trigger determination.

O. Monitoring History

Versioned historical record of monitoring states, signals, and changes.

84. Governance Outputs

PCHM may produce:

Governed Prediction Change Monitoring State Monitoring Reference State Prediction Monitoring Profile
Change Signal Record Change Observation Prediction Basis Change Indicator Prediction Basis Decay
Indicator Assumption Change Indicator Probability-Relevant Change Indicator Uncertainty Change
Indicator Confidence-Relevant Change Indicator Feasibility-Relevant Change Indicator Intervening Event
Record Environmental Drift Indicator Structural Change Indicator Collective Prediction Change Indicator
Predictor–Reality Proximity Change Record Prediction Feedback Signal Monitoring Coverage State
Monitoring Blind Spot Indicator Cumulative Change Profile Monitoring Failure Indicator Change Escalation
Candidate Historical Monitoring Record

The primary output is:

Governed Prediction Change Monitoring State

Potentially material change proceeds to:

RTM — Review Trigger Module

85. Critical Distinctions

Monitoring Without Prediction Identity ≠ Governed Prediction Monitoring

Everything That Changes ≠ Relevant Predictive Change

Higher Monitoring Frequency ≠ Better Governance by Default

Change Signal ≠ Material Change

Signal Relevance ≠ Signal Materiality

Observed Change Without Predictive Association ≠ Governed Change Determination

Prediction Basis Age ≠ Prediction Basis Decay by Default

Same Source Name ≠ Same Source Condition

Replacement Source ≠ Equivalent Source by Default

Change Detected ≠ Probability Automatically Changed

Stable Probability ≠ Stable Confidence

Conditional Prediction ≠ Permanently Valid Conditional Structure

Intervening Event ≠ Material Predictive Change by Default

Approaching Prediction Horizon $\neq$ Automatic Revision Requirement

Gradual Change $\neq$ Non-Material Change

Same Collective Number $\neq$ Same Collective Structure

Stable Aggregate $\neq$ Stable Influence Structure

High Predictor–Reality Proximity $\neq$ Misconduct

Prediction Influence $\neq$ Outcome Control

Prevented Outcome $\neq$ Automatically Failed Prediction

Communication Change $\neq$ Prediction Change

Reliance Change $\neq$ Prediction Change

Use Boundary Change $\neq$ Prediction Revision

More Signals $\neq$ More Governance Events

Multiple Reports $\neq$ Multiple Changes

Multiple Signals $\neq$ Independent Evidence of Change

No New Signal $\neq$ No Change by Default

Unobserved Condition $\neq$ Stable Condition

Individually Minor Changes $\neq$ Collectively Minor Change

Monitoring Intensification $\neq$ Prediction Revision

Monitoring Failure $\neq$ Prediction Error

Event Time $\neq$ Detection Time

Later Knowledge $\neq$ Earlier Monitoring State

Machine-Readable Monitoring $\neq$ Autonomous Revision Authority

Human Observation $\neq$ Automatic Materiality Determination

AI-Detected Change $\neq$ Governed Revision

Monitoring Authority $\neq$ Revision Authority

Change Detection $\neq$ Revision Decision

86. Enterprise Application

PCHM can govern prediction monitoring across:

enterprise forecasting; supply-chain forecasting; financial forecasting; investment analysis; insurance; manufacturing; predictive maintenance; energy systems; logistics; infrastructure; technology forecasting; public-sector planning; emergency preparedness; AI prediction systems; autonomous systems; AI agents; model ensembles; prediction markets; Human-AI forecasting systems.

Its function remains constant:

make changes surrounding a prediction observable and governable before those changes silently turn a once- legitimate prediction into a stale predictive state.

87. Use Case 1 — Semiconductor Supply Prediction After Export Restrictions

A manufacturer has a Governed Prediction:

72% probability of sufficient semiconductor supply during the next six months.

Its monitored conditions include:

supplier capacity; export availability; shipping capacity; inventory; production commitments.

Three weeks later, a major exporting jurisdiction announces restrictions affecting a principal supplier.

The prediction has not yet changed.

Its displayed probability remains:

72%

PCHM detects:

Change Signal: New export restriction.

Predictive Association: Prediction Basis + feasibility conditions + supply assumption.

Change Direction: Potentially Weakening.

Change Magnitude: Potentially Substantial.

Persistence: Unable to Determine.

Monitoring State: Potentially Material Change.

PCHM creates:

Change Escalation Candidate → RTM

It does not revise 72% to another number.

That distinction is essential:

Material Change Detection ≠ Probability Recalculation

The downstream PMRS process determines whether formal review and revision are required.

88. Use Case 2 — Failure Prediction Changes the Environment Through Preventive Action

An industrial AI system predicts:

89% probability of critical equipment failure within seven days.

The prediction is communicated and relied upon.

Engineers perform preventive maintenance and replace a component.

The system's operational condition materially changes.

PCHM records:

Prediction Use → Preventive Intervention → Changed Operational Reality

The monitored conditions now differ materially from those under which the 89% prediction was formed.

PCHM identifies:

operational-condition change; prediction feedback; Self-Negating Dynamics; changed feasibility of the original failure pathway.

The prediction is not retrospectively classified as wrong.

PCHM instead creates a governed record showing that the environment changed after prediction formation.

Prediction Influence ≠ Predictive Accuracy

Prevented Outcome ≠ Automatically Failed Prediction

Changed Operational Reality ≠ Historical Prediction Rewrite

This record can later support PMRS review and POFPS performance interpretation without destroying the historical integrity of the original prediction.

89. Canonical Boundary

PCHM governs observation, association, comparison, classification, and preservation of changes potentially affecting an existing Governed Prediction, including Prediction Basis Decay, assumption change, probability- and uncertainty-relevant change, feasibility conditions, collective structure, Predictor–Reality Proximity, Operational Conditions, external conditions, intervening events, Prediction Horizon, prediction-use feedback, reflexive change, environmental drift, Monitoring Coverage, Monitoring Blind Spots, cumulative change, and Change Escalation Candidates. It does not determine final materiality for Prediction Review, create the Review Trigger, recalculate probability, revise the prediction, supersede or withdraw the prediction, determine Operational Reality, or assess predictive performance.

PCHM owns:

Prediction Change Monitoring

RTM owns:

Review Trigger determination

PRVM owns:

Prediction Revision

PSSM owns:

Prediction Supersession

PWM owns:

Prediction Withdrawal

90. Canonical Closing Statement

Predictions rarely become stale because time alone passes; they become stale because the world changes while the prediction remains still.

PCHM makes those changes visible without confusing observation with judgment, signal with materiality, or monitoring with revision.

A prediction cannot govern its changing future if it cannot detect that the conditions around it have changed.

It

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contact@originopenfoundation.org